56-Mark

# University of Information Technology & Sciences (UITS)

## Faculty of Science and Engineering

## Department of Computer Science and Engineering

**Program: B.Sc. in CSE**

**Term Final Examination, Autumn 2025**

**Course Title: Coordinate Geometry Linear Algebra and Vector Analysis**

**Course Code: MATH0541211**

**Marks: 50** | **Time: 3*(three)* hours**

**(Answer all questions)**

| Q.No. | Questions | Marks |
|---|---|---|
| 1. | Define consistent and inconsistent matrix. Solve the following system of linear equations: $$\begin{aligned} x_1+x_2-3x_3-4x_4&=-1\\ 2x_1+2x_2+2x_3-3x_4&=2\\ 2x_1+x_2+5x_3+x_4&=5\\ 3x_1+6x_2-2x_3+x_4&=8\end{aligned}$$ 2, 1, 5, 0, 4/5 | [10] |
| 2. | a) Define Idempotent matrix. Demonstrate that $A=\begin{pmatrix} 2 & -2 & -4\\ -1 & 3 & 4\\ 1 & -2 & -3\end{pmatrix}$ is Idempotent. | [04] |
| | b) Solve the following linear equations using Inverse Matrix Method: $$\begin{aligned} 4x-2y+z&=16\\ 6x-y-4z&=0\\ 2x+y-2z&=8\end{aligned}$$ 1, 0, 8 | [06] |
| 3. | a) Define Plane. Compute the equations to the planes which is perpendicular to the plane $2x-y+3z-7=0$ and passing through the points (0,0,0) and (2,0,−4). $2x+7y+2=0$ | [05] |
| | b) Direction cosine of two straight lines are connected by the relations $2l+2m-n=0$ and $mn+nl+lm=0$. Show that the lines are perpendicular to each other. 1/3, 2/3, −2/3 | [05] |
| 4. | a) Discover the constant a b and c so that a irrotational vector **V** is given by $\boldsymbol{V}=(4x+3y-az)\hat{\imath}+(bx-5y+z)\hat{\jmath}+(x+cy+z)\hat{k}$. A, 3, 1 | [03] |
| | b) Suppose $\boldsymbol{A}=2yz\hat{\imath}-x^2y\hat{\jmath}+xy^2\hat{k}$, $\boldsymbol{B}=xz\hat{\imath}+yz\hat{\jmath}+xy\hat{k}$ and $\emptyset=2z^2xy^3$. Obtain $(A\times\nabla)\emptyset$ and $(B.\nabla)A$ at the point $P(1,0,1)$. 0, 1, 0 | [04] |
| | c) Estimate the projection of $\boldsymbol{A}=4\hat{\imath}-\hat{\jmath}+2\hat{k}$ on the vector $\boldsymbol{B}=\hat{\imath}+\hat{\jmath}+\hat{k}$. | [03] |

$6/\sqrt{3}$

5. a) Suppose $\mathbf{A}= x^2z^2\hat{i}-2y^2z^2\hat{j}+xy^2z\hat{k}$. Find curl curl $\mathbf{A}=\nabla\times(\nabla\times A)$. [04]

b) Show that $\nabla r^n = nr^{n-2}\mathbf{r}$. [03]

c) Estimate the unit tangent vector to any point at $t = 4$ on the curve $x=t^2-t,\ y=4t-3,\ z=2t^2-8t$. [03]

$7i+9j+8k$

$\sqrt{129}$